

LSTM

Multi-label toxic comment classification · Evaluated on 15,958 held-out samples · Generated 2026-09-25

Target: Weighted-F1 ≥ 0.75

PASS

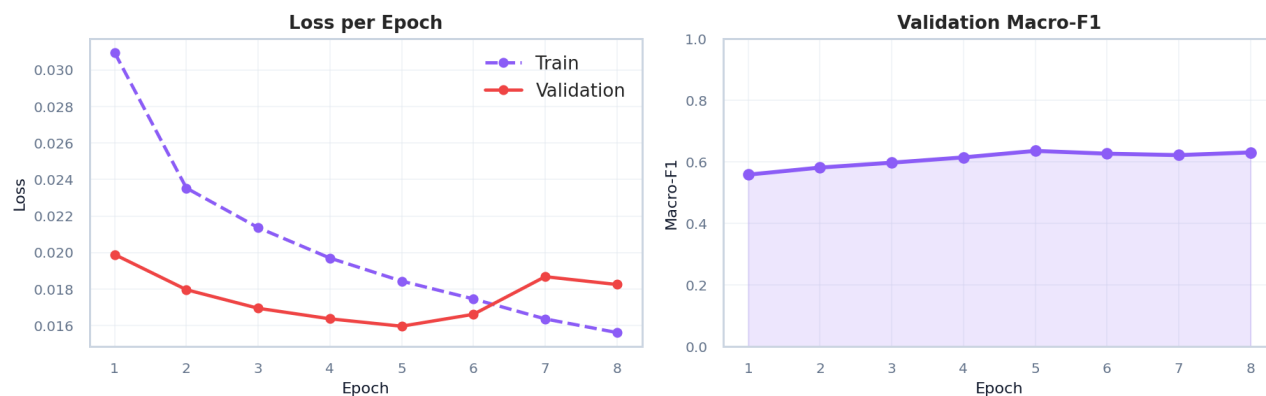
WEIGHTED-F1	MACRO-F1	MICRO-F1	MACRO-AUC
0.7643	0.6368	0.7559	0.9571

Headline Numbers

- **Exact-match accuracy:** 91.63% of comments have all six labels predicted correctly.
- **Micro precision / recall:** 0.7357 / 0.7773.
- **Best class:** obscene — F1 0.8251, AUC 0.9575.
- **Hardest class:** threat — F1 0.4219, support 46.

Training Dynamics

Loss and macro-F1 across training epochs. Early stopping selects the epoch with the best *untuned* validation macro-F1. A widening train–validation gap indicates overfitting; a plateau in validation F1 signals convergence.

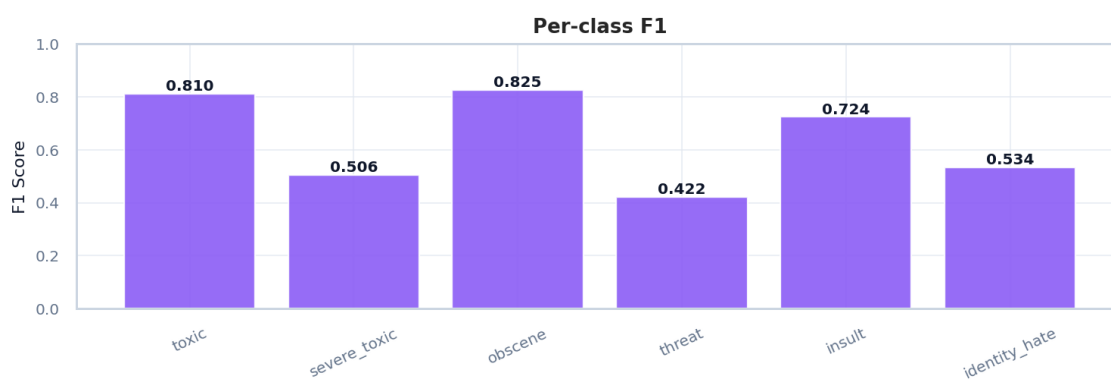


Best epoch: 5 of 8 · peak val macro-F1 = 0.6355.

Per-Class Performance

Each label uses its own decision threshold, tuned on a separate calibration split to maximise macro-F1. The toxic class additionally gates the other five labels: rows with $P(\text{toxic})$ below the gate are forced to 0.

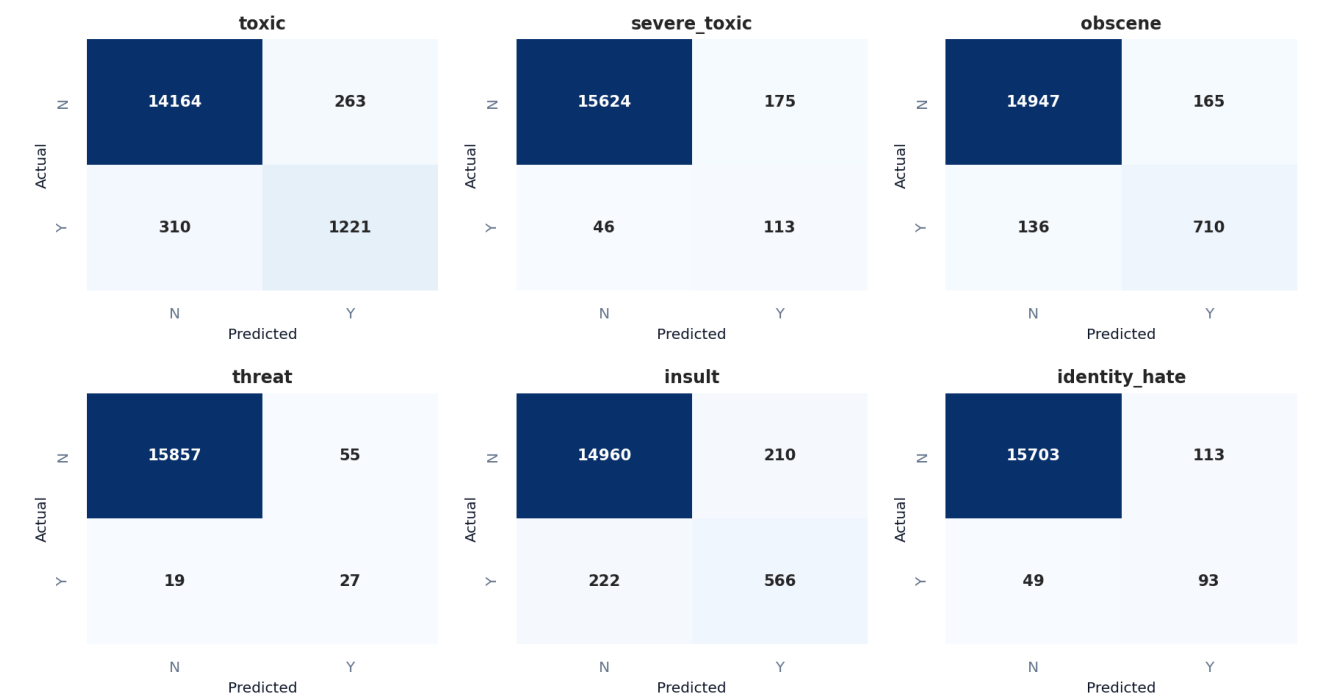
Label	F1	Precision	Recall	AUC	Support
toxic	0.8100	0.8228	0.7975	0.9822	1,531
severe_toxic	0.5056	0.3924	0.7107	0.9913	159
obscene	0.8251	0.8114	0.8392	0.9575	846
threat	0.4219	0.3293	0.5870	0.9371	46
insult	0.7238	0.7294	0.7183	0.9372	788
identity_hate	0.5345	0.4515	0.6549	0.9375	142



Confusion Matrices

Counts of correct and incorrect predictions per class. Colors are scaled per class so that minority classes remain readable.

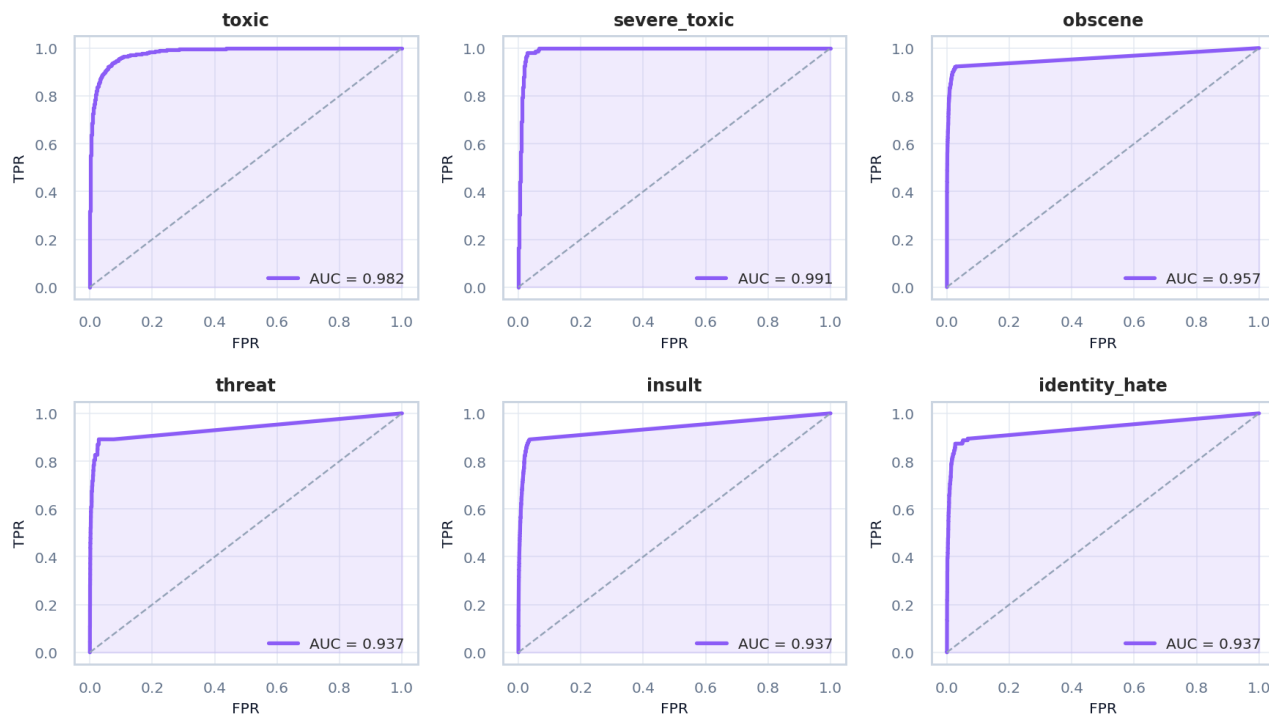
LSTM — Confusion Matrices



ROC Curves

ROC curves show ranking quality. AUC near 1.0 means the model ranks positives above negatives reliably, even when a class is very rare (e.g. *threat* is <0.3% of rows).

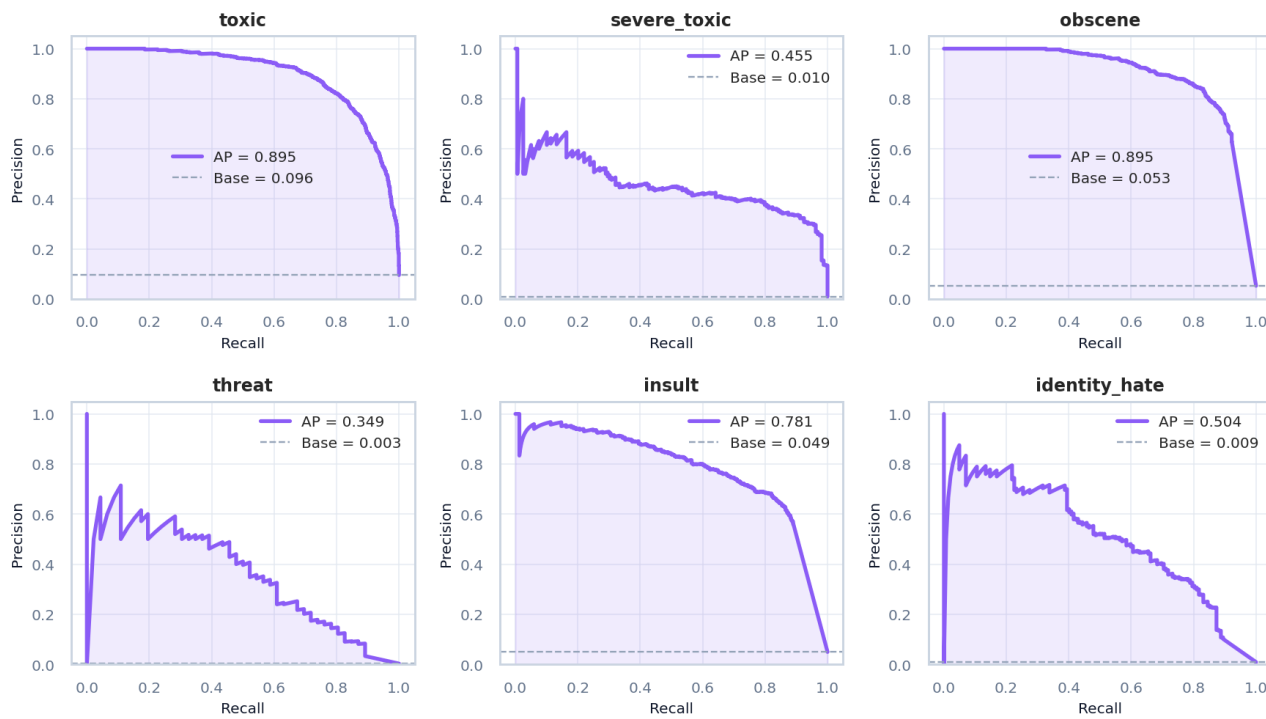
LSTM — ROC Curves



Precision-Recall Curves

PR curves are more informative than ROC under heavy class imbalance. The dashed line is the class base rate; the area above it is the model's lift.

LSTM — Precision-Recall Curves



Model Configuration

Parameter	Value
Architecture	LSTM — bidirectional, 2 layers, hidden=160
Embedding	GloVe-6B 300d (frozen 1 epoch, then fine-tuned)
Pooling	Attention + max + mean (concatenated)
Classifier	MLP 960 → 320 → 6 with dropout 0.35
Loss	Focal loss ($\alpha=0.25$, $\gamma=2.0$) + pos_weight
Optimizer	AdamW (embed_lr=2e-5, base_lr=2e-4, wd=5e-4)
Batch size	64
Max sequence	220 tokens
Epochs trained	8
Early stopping	Untuned macro-F1 on val (patience=3)
Evaluated on	15,958 samples

Decision Thresholds (macro-F1 tuned)

Label	Threshold
toxic	0.400
severe_toxic	0.250
obscene	0.400
threat	0.200
insult	0.450
identity_hate	0.250
toxic gate (t_toxic)	0.550

Methodology Notes

- Multi-label problem (6 independent binary labels); focal loss ($\alpha=0.25$, $\gamma=2.0$) with per-class pos_weight handles severe class imbalance.
- GloVe-6B 300d embeddings initialise the embedding layer; the layer is frozen for the first epoch, then fine-tuned.
- Model selection uses UNTUNED validation macro-F1 to avoid leaking threshold choices into early stopping.
- Per-class thresholds and the toxic gate are tuned on a separate calibration split, never on the held-out test set.